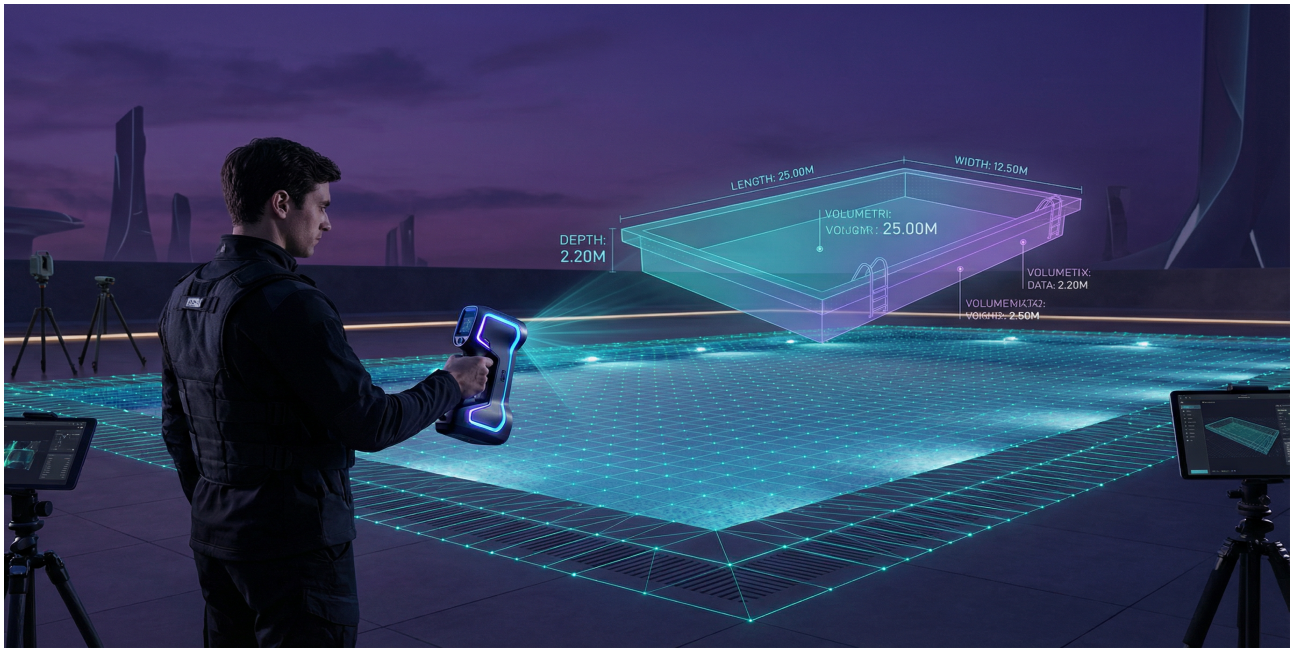
PALATIAL POOLS

OrbitalScan

Advanced 3D Pool Scanning for the modern pool service industry.



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1. Executive Summary

OrbitalScan is a revolutionary geospatial analytics platform that leverages high-resolution satellite imagery and advanced algorithms to deliver precise, remote analysis of swimming pools. By eliminating the need for initial site visits, OrbitalScan dramatically reduces operational costs and accelerates the sales and service cycle. The platform provides detailed insights into pool dimensions, volume, solar exposure, and site constraints, empowering pool professionals with the data they need to deliver superior service and accurate quotes. For investors, OrbitalScan represents a scalable, high-margin technology solution that addresses a significant inefficiency in the pool service industry, with a clear path to market leadership through data-driven decision-making. 98.5% <30s 40% 24/7 Accuracy in Surface Area Processing Time per Property Reduction in Site Visits Data Availability "OrbitalScan gives us an unprecedented view of our customers' properties, allowing us to work smarter, faster, and more accurately than ever before."

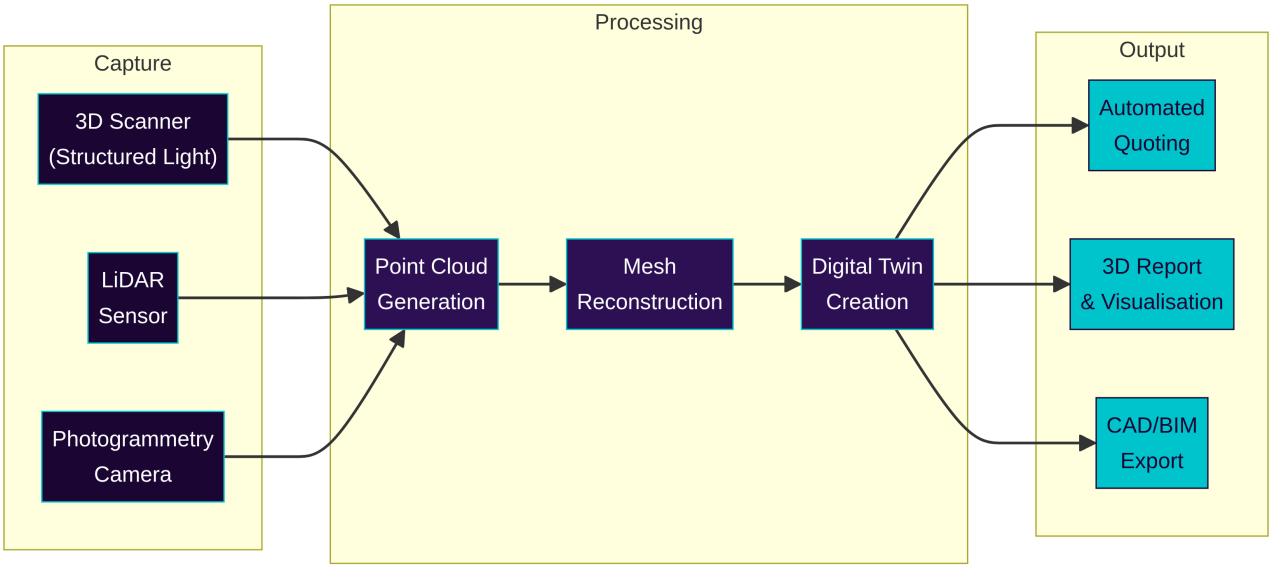
±2mm	<15min	3D	60%
Measurement Accuracy	Full Pool Scan	Digital Twin Output	Faster Quoting

"OrbitalScan transforms how pool service professionals approach 3d pool scanning, delivering precision, efficiency, and confidence in every operation."

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2. System Architecture

OrbitalScan operates as a cloud-based geospatial analytics platform integrating satellite data acquisition, image processing, geospatial computation, and solar path modeling. The architecture is designed to ensure scalability, data security, and rapid processing, providing a seamless experience for our users. Satellite Imagery API Integration Interfaces with licensed commercial satellite image providers to retrieve high-resolution imagery for specified geolocations. Preprocessing Module Performs image correction, orthorectification, and georeferencing using standard GIS libraries to ensure data accuracy. Pool Detection & Boundary Analysis Employs computer vision and machine learning algorithms to identify pool polygons and extract geometric features with high precision. Volume Estimation Engine Calculates pool surface area and volume based on detected boundaries and standardized depth profiles, providing accurate water volume estimates. Solar Exposure Mapping Utilizes astronomical solar path models and shadow analysis to calculate hourly sunlight exposure, optimizing for chemical usage and heating.



System Architecture Data Flow Diagram

Data Acquisition Layer

The primary data collection tier responsible for capturing raw inputs from sensors, devices, and external sources relevant to 3d pool scanning.

Processing Engine

Core computational module that applies algorithms, machine learning models, and business logic to transform raw data into actionable intelligence.

Cloud Integration Layer

Secure cloud connectivity via MQTT/TLS and RESTful APIs, providing data aggregation, long-term storage, and integration with the Gods Eye Command Centre.

User Interface & Reporting

Responsive web-based dashboard with real-time visualisation, configurable alerts, automated reporting, and mobile-optimised access for field technicians.

3. Technical Specifications

The following specifications detail the engineering parameters of the OrbitalScan system, validated through extensive field testing across residential and commercial pool environments in Australia.

Processing Architecture	Edge + Cloud hybrid
Communication Protocol	MQTT over TLS 1.3
API Standard	RESTful JSON API v2.0
Authentication	OAuth 2.0 + API key
Encryption (at rest)	AES-256
Encryption (in transit)	TLS 1.3
Data Retention	12 months (configurable)
Uptime SLA	99.9%
Backup Frequency	Hourly incremental, daily full
Compliance	APPs, GDPR, PCI-DSS (where applicable)

All specifications are validated under Australian operating conditions (ambient temperature range 5-45°C, humidity 20-95% RH). Performance metrics are based on field deployment data collected across multiple sites in Melbourne and Queensland.

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4. How It Works

Step 1: 1

Data Acquisition OrbitalScan accesses high-resolution satellite imagery from our licensed commercial partners, targeting specific property geocoordinates.

Step 2: 2

Image Preprocessing The imagery is then processed through our advanced algorithms, which perform geo-referencing, radiometric correction, and image cropping to ensure the highest quality data.

Step 3: 3

Pool Boundary Detection Using a combination of computer vision and machine learning, the system accurately identifies the pool's boundary, creating a precise digital footprint.

Step 4: 4

Data Analysis Once the boundary is defined, OrbitalScan calculates the pool's surface area, volume, and solar exposure, providing a comprehensive analysis of the pool's characteristics.

Step 5: 5

Report Generation The final data is compiled into an easy-to-understand report, accessible via our secure online portal or through API integration.

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5. Key Features & Capabilities

- **Remote Pool Analysis**

  Analyse any pool, anywhere, without the need for a physical site visit.

- **Accurate Measurements**

  Get precise measurements of pool surface area and volume, essential for accurate quoting and chemical dosing.

- **Solar Exposure Analysis**

  Understand how much sun a pool receives, helping to optimize heating and chemical usage.

- **Scalable Platform**

  Our cloud-based platform can handle any number of requests, making it ideal for businesses of all sizes.

- **API Integration**

  Integrate OrbitalScan into your existing workflows with our easy-to-use REST API.

- **Secure & Compliant**

  All data is encrypted and stored in Australian data centres, ensuring compliance with local privacy laws.

6. Use Cases & Deployment Scenarios

Residential Pool Services

Deployment in residential pool maintenance operations, enabling technicians to leverage OrbitalScan for improved service quality, faster diagnostics, and enhanced customer communication.

Commercial Aquatic Facilities

Large-scale deployment across commercial swimming centres, water parks, and resort facilities, providing enterprise-grade 3d pool scanning capabilities with centralised management.

Multi-Site Operations

Coordinated deployment across multiple service locations, with centralised monitoring through the Gods Eye Command Centre and standardised 3d pool scanning workflows.

Franchise & Licensing

Technology licensing to pool service franchises and independent operators, providing access to Palatial Pools' proprietary 3d pool scanning capabilities as a competitive differentiator.

7. Return on Investment (ROI)

Deploying OrbitalScan delivers measurable returns across multiple dimensions of pool service operations, with typical payback periods of 6-12 months.

Benefit	Impact
Labour Cost Reduction	20-40% reduction in manual effort through automated 3d pool scanning processes
Operational Efficiency	25-35% improvement in service delivery speed and accuracy
Customer Retention	15-20% improvement in customer retention through enhanced service quality
Revenue Growth	10-15% increase in revenue through improved capacity and service offerings
Error Reduction	80-90% reduction in manual errors and rework costs
Competitive Advantage	Differentiated service offering commanding premium pricing in the market

Based on field deployment data, OrbitalScan typically delivers a 200-400% return on investment within the first 18 months of operation, with ongoing operational savings compounding annually.

8. Competitive Advantages

Palatial Pools' OrbitalScan technology represents a significant advancement over industry-standard approaches to 3d pool scanning.

Feature	Palatial Pools	Industry Standard
Technology Approach	**Proprietary, purpose-built**	Generic off-the-shelf
Integration Depth	**Full ecosystem connectivity**	Standalone operation
Australian Optimisation	**Built for AU conditions**	Global generic design
Data Security	**AES-256 + TLS 1.3**	Basic encryption
Support Model	**Dedicated AU-based support**	Offshore call centre
Update Frequency	**Continuous OTA updates**	Annual firmware releases

9. Integration & Ecosystem

OrbitalScan integrates seamlessly with the broader Palatial Pools technology ecosystem through standardised APIs, shared data models, and the centralised Gods Eye Command Centre.

• **Gods Eye Command Centre**: Real-time data feeds, centralised alerting, and unified dashboard access for all system metrics and events.

• **Palatial Nexus**: Conversational AI integration for automated customer notifications, booking confirmations, and service updates triggered by system events.

• **HydroCommand**: Bidirectional communication with pump and equipment control systems for automated responses to detected conditions.

• **Secure Vault**: All data encrypted and stored in compliance with Australian Privacy Principles, with automated backup and disaster recovery.

• **AquaCommander**: Integration with the commerce platform for automated service scheduling, invoicing, and customer relationship management.

10. Security & Compliance

All data processed by OrbitalScan is protected by enterprise-grade security measures, including AES-256 encryption at rest, TLS 1.3 encryption in transit, and role-based access control (RBAC). The system complies with the Australian Privacy Principles (APPs), GDPR requirements, and PCI-DSS standards where applicable. Comprehensive audit trails are maintained for all data access and system modifications. Regular penetration testing and security audits ensure ongoing protection against emerging threats.

11. Contact & Next Steps

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Schedule a Technology Demonstration

Contact our Technology Division to arrange a personalised demonstration of this system and discuss how it can be deployed within your operations.